

INTRODUCTION TO DATA MINING

Key Concepts and System Overview

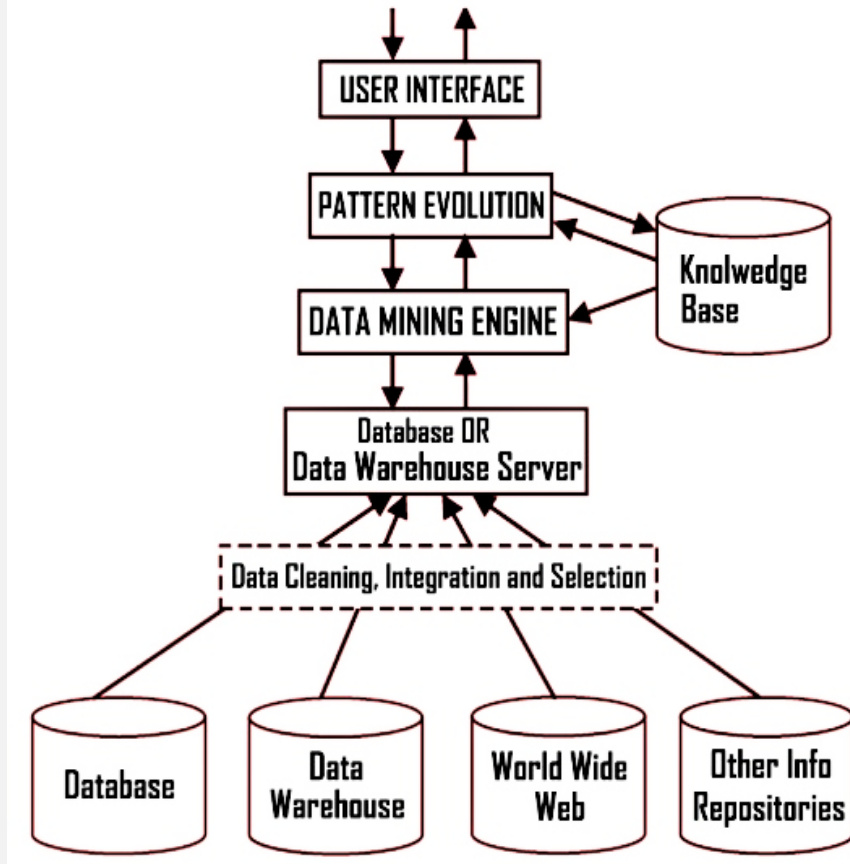
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RELATION TO STATISTICS

- Data mining draws heavily from statistics for pattern identification.
- It uses statistical models to discover relationships and trends in large datasets.

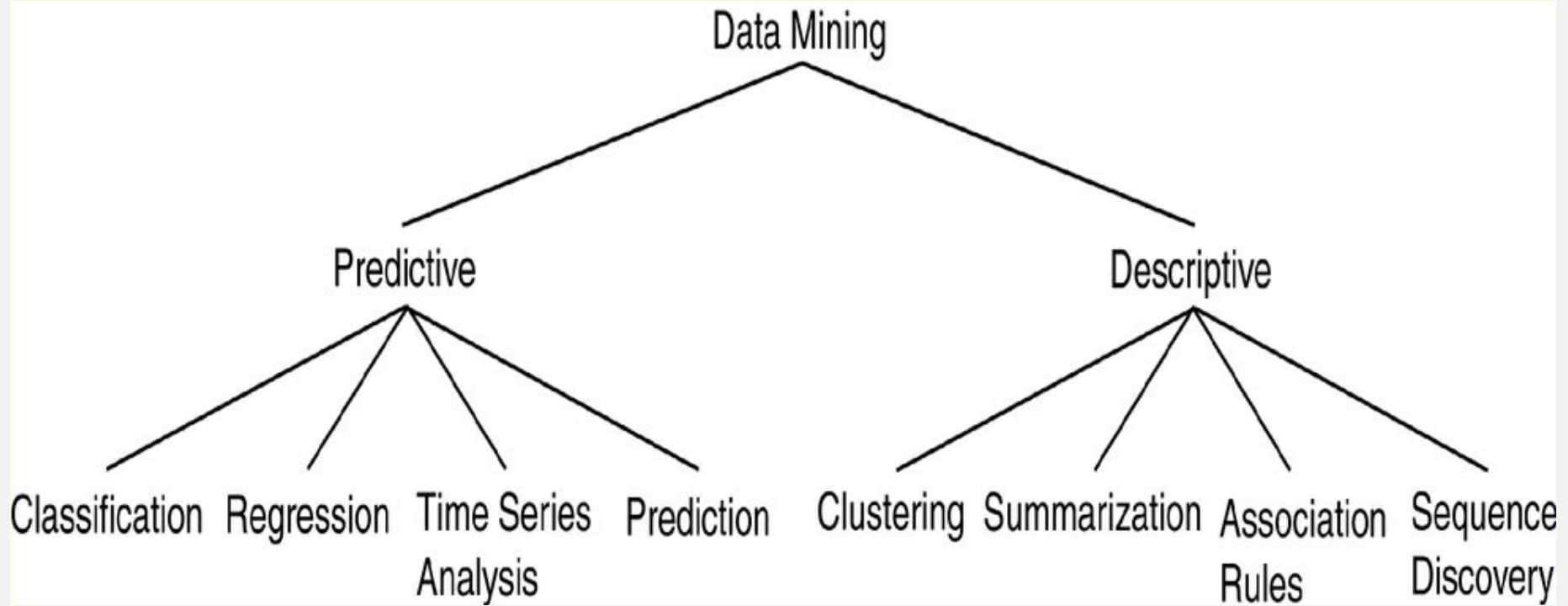


DATABASES



- Databases store large volumes of structured data efficiently.
- They serve as the foundation for data mining by enabling fast data retrieval and management.

DATA MINING FUNCTIONALITIES



- Characterization and discrimination
- Association and correlation analysis
- Classification and prediction
- Clustering and outlier analysis

STEPS IN DATA MINING PROCESS

- 1. Data cleaning
- 2. Data integration
- 3. Data selection
- 4. Data transformation
- 5. Data mining
- 6. Pattern evaluation
- 7. Knowledge presentation

STEPS IN DATA MINING PROCESS

Data Cleaning

- Definition: Removing errors, inconsistencies, and missing values from raw data.
- Example: In a rehabilitation hospital, a patient database may have entries like missing age or incorrectly spelled diagnosis. During data cleaning, blanks are filled or corrected (“Paraliss” → “Paralysis”; missing age imputed with an average).

Data Integration

- Definition: Combining data from multiple sources into a unified view.
- Example: Merging electronic health records from physical therapy, occupational therapy, and speech therapy departments into a single comprehensive database for deeper analysis.

STEPS IN DATA MINING PROCESS

Data Selection

- Definition: Choosing the relevant data needed for a specific analysis from the integrated dataset.
- Example: Selecting only patients with stroke rehabilitation records in the last year for a study about therapy outcomes.

Data Transformation

- Definition: Converting selected data into a suitable format for mining (e.g., normalization, aggregation).
- Example: Transforming heights from centimeters to meters; categorizing age into groups like "Child," "Adult," and "Senior."

STEPS IN DATA MINING PROCESS

Data Mining

- Definition: Applying algorithms to extract meaningful patterns or relationships from data.
- Example: Using clustering algorithms to group patients by similar rehabilitation progress patterns, or association rules to find which therapies are often prescribed together.

Pattern Evaluation

- Definition: Assessing and selecting the most significant and useful patterns discovered in data mining.
- Example: Reviewing the groups identified to check if they truly reveal new, actionable insights —like discovering a subgroup of stroke patients who improve faster with a multi-disciplinary approach.

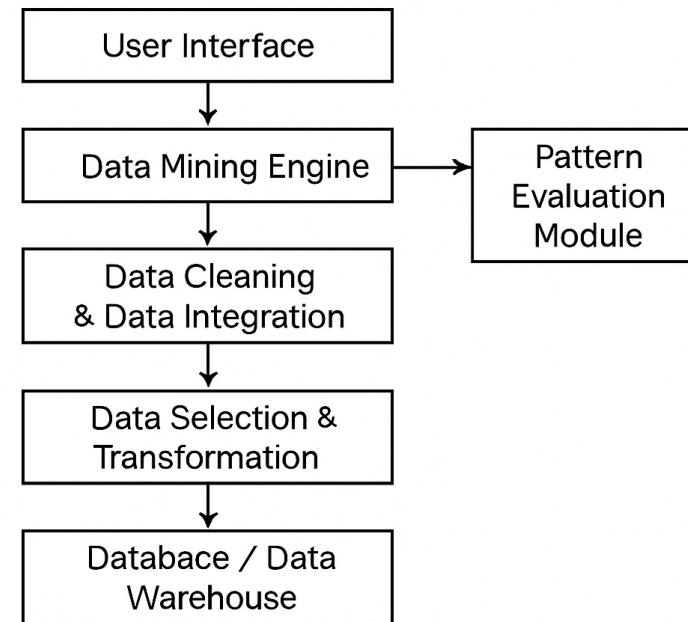
Knowledge Presentation

- Definition: Presenting mined insights in understandable formats like charts, reports, or dashboards.
- Example: Creating visual graphs to show therapy outcome improvements over time, shared with the rehab unit's staff for decision-making.

ARCHITECTURE OF TYPICAL DATA MINING SYSTEMS

- Database/data warehouse server
- Data mining engine
- Pattern evaluation module
- Graphical user interface
- Knowledge base

Architecture of a Typical Data Mining System



CLASSIFICATION OF DATA MINING SYSTEMS

- Based on the type of data mined (e.g., relational, transactional)
- Based on the kind of knowledge discovered (e.g., clusters, rules)
- Based on the techniques used (e.g., machine learning, statistics)
- Based on the architecture used (eg: Loose, Semi-tight, Tight Coupling)

OVERVIEW OF DATA MINING TECHNIQUES

- Classification (Decision trees, Naive Bayes)
- Clustering (k-means, hierarchical clustering)
- Association rule mining (Apriori, FP-Growth)
- Regression and prediction methods

ARCHITECTURE

1. Type of Data Mined
2. Type of Knowledge Mined
3. Type of Techniques Used
4. Type of Architecture Used

1. CLASSIFICATION BASED ON TYPE OF DATA MINED

- Relational Databases (SQL, Oracle)
- Data Warehouses
- Transactional Data
- Spatial & Temporal Data
- Multimedia, Text, and Web Data
- Example: Web mining from search logs

2. CLASSIFICATION BASED ON TYPE OF KNOWLEDGE MINED

- Concept/Class Description
- Association Rules
- Classification & Prediction
- Clustering
- Outlier Detection
- Evolution & Trend Analysis

3. CLASSIFICATION BASED ON TECHNIQUES USED

- Database-Oriented
- Machine Learning
- Statistical Methods
- Visualization Techniques
- Pattern Recognition
- Example: Apriori (DB-based), Decision Trees (ML)

4. CLASSIFICATION BASED ON ARCHITECTURE USED

- No Coupling: No DBMS integration
- Loose Coupling: DB used for data retrieval
- Semi-Tight Coupling: Some DB functions used
- Tight Coupling: Fully integrated with DBMS
- Example: SQL Server Analysis Services (tight)

SUMMARY: CLASSIFICATION TYPES

- Data: Relational, Web, Multimedia
- Knowledge: Classification, Clustering, Associations
- Techniques: ML, Stats, Visualization
- Architecture: No, Loose, Semi-tight, Tight Coupling

CASE STUDY: HEALTHCARE DATA MINING

- Objective: Predict patient readmission risk using historical health records.
- Dataset: 100,000+ hospital discharge records.
- Method: Classification using Decision Trees and Logistic Regression.
- Outcome: Improved discharge planning and reduced readmission rates by 15%.
- Benefit: Enhanced patient care and cost savings for hospitals.

ALGORITHM EXAMPLE: DECISION TREE

- Used for classification tasks.
- Example: Predicting heart disease based on age, blood pressure, cholesterol.
- Steps:
 1. Select the best attribute using a measure like Information Gain.
 2. Split the dataset based on the selected attribute.
 3. Repeat recursively for each split subset.
- Tools: Implemented in WEKA, R, Python (scikit-learn).

ALGORITHM EXAMPLE: K-MEANS CLUSTERING

- Used for grouping similar data points.
- Example: Segmenting patients into risk categories.
- Steps:
 1. Choose k initial centroids.
 2. Assign each data point to the nearest centroid.
 3. Recalculate centroids and repeat until convergence.
- Tools: Available in WEKA, R (cluster package), Python (sklearn).